Designing Microsoft Azure Infrastructure Solutions: AZ:305

TECHNOLOGY

Design a Solution for Relational Data

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Learning Objectives

By the end of this lesson, you will be able to:

- Gain insights into Azure SQL database to simplify database management
- Identify the deployment models in SQL Server to provide versatile solutions for diverse application needs
- List the Database Service Tier Sizing to ensure optimal performance and cost efficiency for the database
- Assess a scalable database solution to ensure a scalable and sustainable database infrastructure
- Gain insights into Azure SQL Edge to create a high-performance data storage and processing layer for IoT applications

TECHNOLOGY

Selecting an Appropriate Data Platform Based on Requirements

Azure SQL Database

It is a fully managed platform as a service (PaaS) database engine.

Features

- Provides high-performance, reliability, complete management, and security
- Supports multiple programming languages
- Migrates users' existing SQL Server databases using the Azure Database Migration Service

Azure SQL Database: Benefits

- Simplifies database management
- Reduces the need for extensive code
- Facilitates seamless synchronization and migration of SQL Server-stored data
- Ensures uninterrupted business operations by overcoming technology constraints

Azure SQL Database

It offers the following deployment options:

01

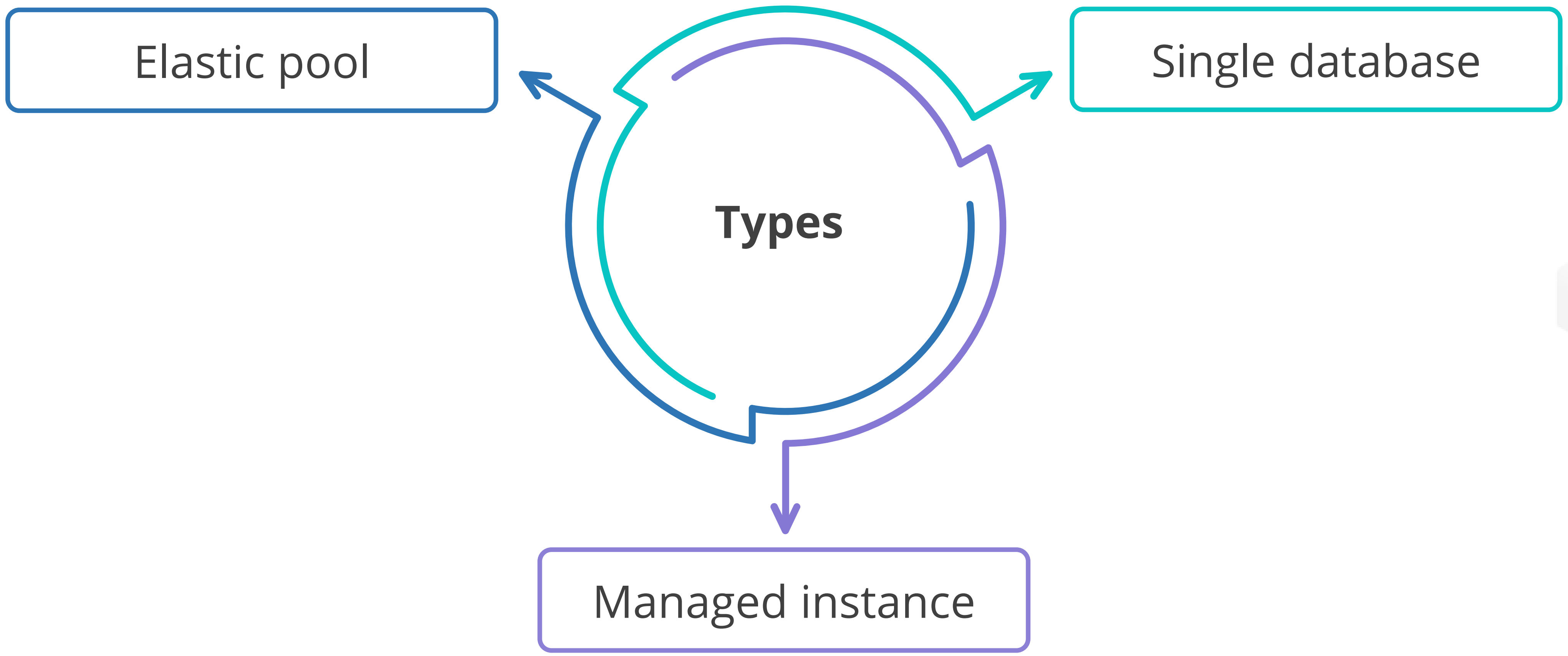
As a single database with its own set of resources managed via a logical server. A single database is similar to a contained database in SQL Server.

02

An elastic pool, which is a collection of databases with a shared set of resources managed via a logical server

Deployment Models

These models in SQL Server provide versatile solutions for diverse application needs.



Deployment Models

Single database

It is maintained by an SQL Database server and deployed on an Azure VM; it is the simplest deployment method.

Elastic pool

It is a collection of interconnected databases that pool resources; individual databases can be added and removed from an elastic pool.

Managed instance

It is a fully managed database instance that makes migrating on-premise SQL databases simple.

Deploy Azure SQL Database

The five sections in the Azure portal to be filled out during deployment for Azure SQL Database and Azure SQL Managed Instance are:

Basics

- Subscription
- Resource group
- Managed instance / server name
- Database name (DB)
- Admin login
- Password
- Region
- Opt-in for pools (DB)
- Compute + storage

Network connectivity

- Public vs private access
- VNet / firewall rules
- Connection type (MI)

Additional settings

- Data source (DB)
- Server collation (MI)
- Database collation (DB)
- Time zone (MI)
- Opt-in for Azure Defender (DB)

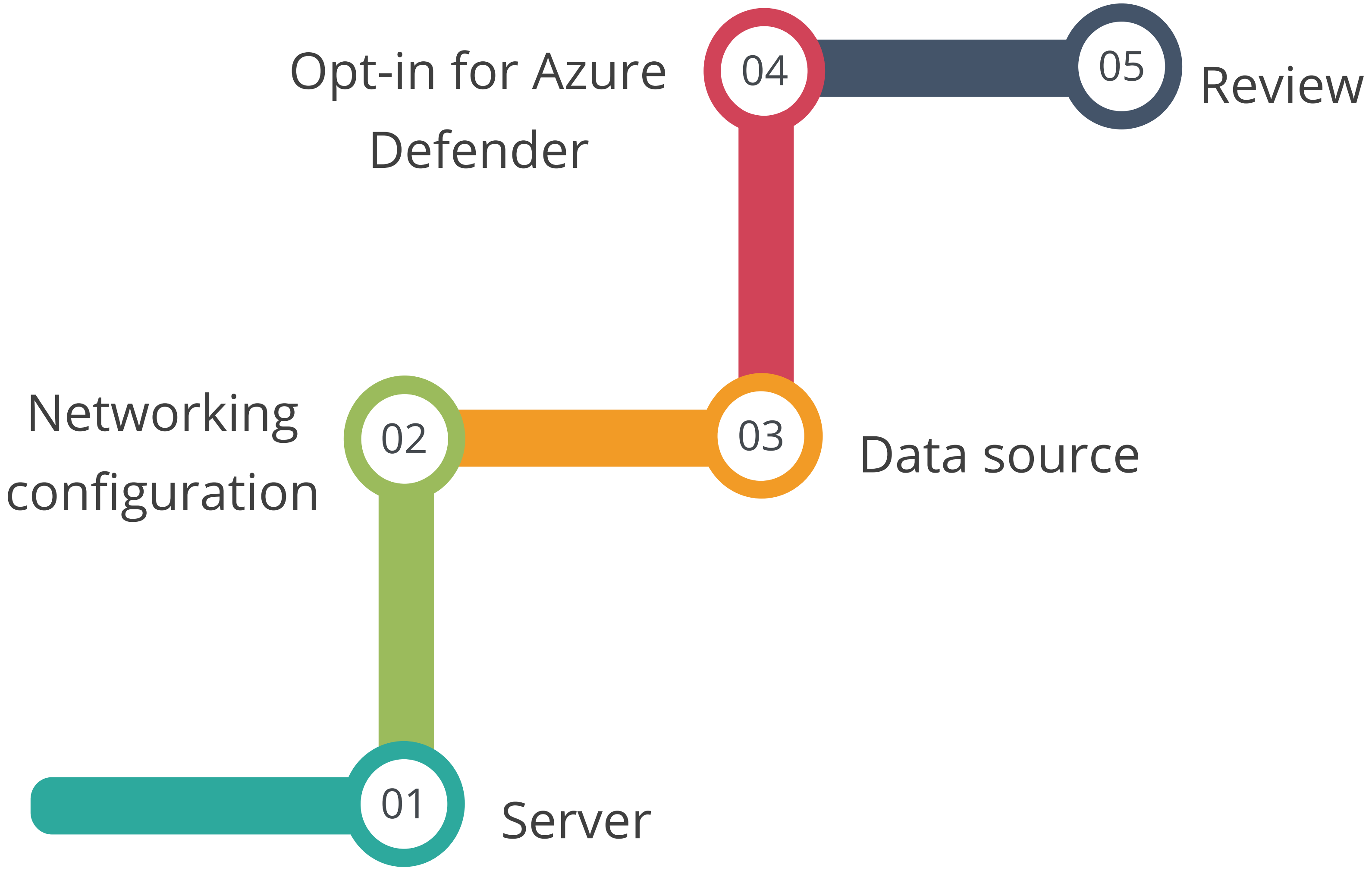
Tags (DB)

Review and create

- Terms and privacy

Deploy Azure SQL Database

The five options available for sections in the Azure portal are:

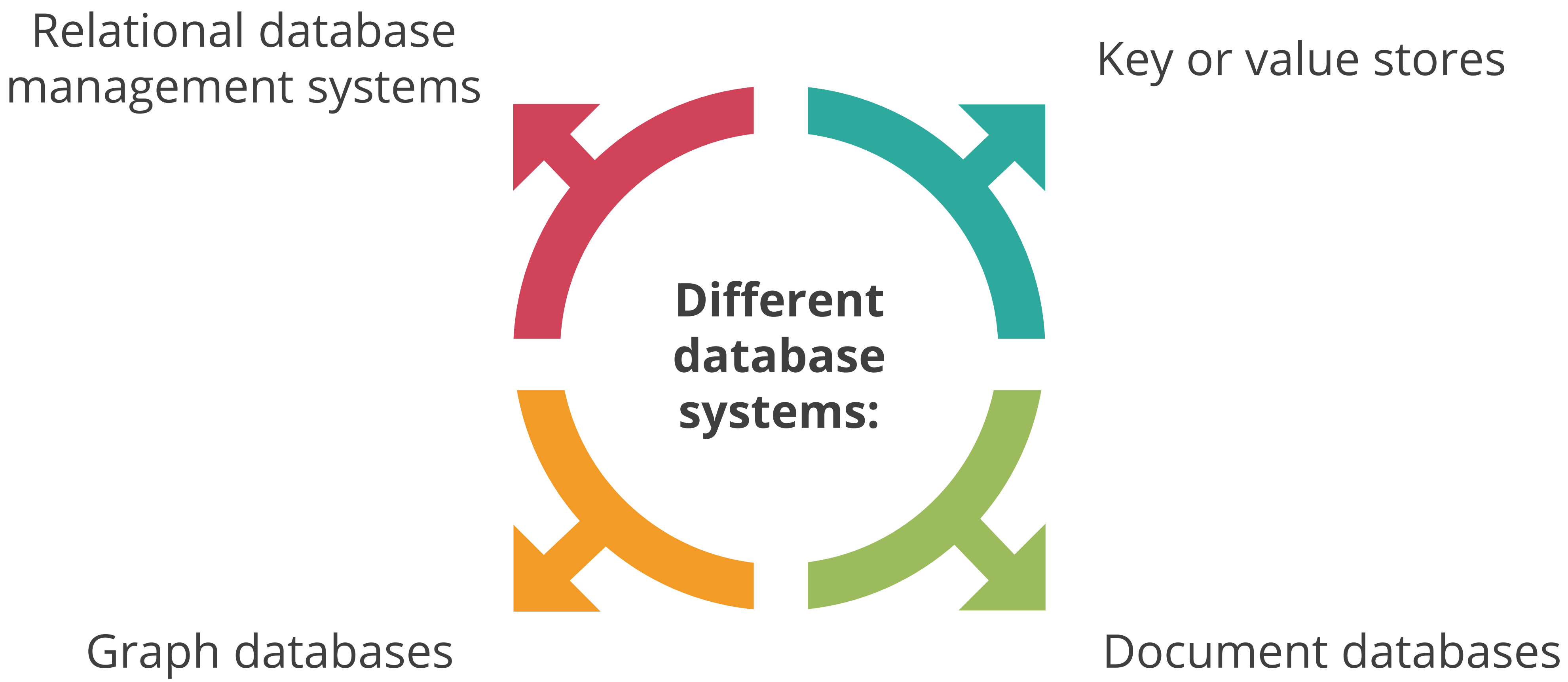


Deploy Azure SQL Database

Server	The name of the server must be unique across Azure.
Networking configuration	A public endpoint can be allowed to link to a controlled instance over the internet.
Data source	The user deploys the instance first and then the databases within it.
Opt-in for Azure Defender	After the instance has been deployed, the user can allow Azure Defender.
Review	The user can check the deployment choices in the Review and Create section.

Recommend the Right Data Store

The different options available among SQL and NoSQL databases include:



Relational Database Management Systems (RDBMS)

It organizes data as a series of two-dimensional tables with rows and columns.

Each table has its own columns, and every row in a table has the same set of columns.

It is based on Structured Query Language (SQL).

It follows the ACID (Atomic, Consistent, Isolated, Durable) model to ensure transactional consistency when updating information.

Key or Value Stores

It is a large hash table.

Opaque to data store

Key	Value
AAAAA	1101001111010100110101111...
AABAB	1001100001011001101011110....
DFA766	0000000000101010110101010...
FABCC4	1110110110101010100101101...

- It associates each data value with a unique key.
- Most key or value stores only support simple query, insert, and delete operations.
- An application can store arbitrary data as a set of values.
- Some key or value stores impose limits on the maximum size of values.

Key or Value Stores

Services of key or value stores are:

Document Databases

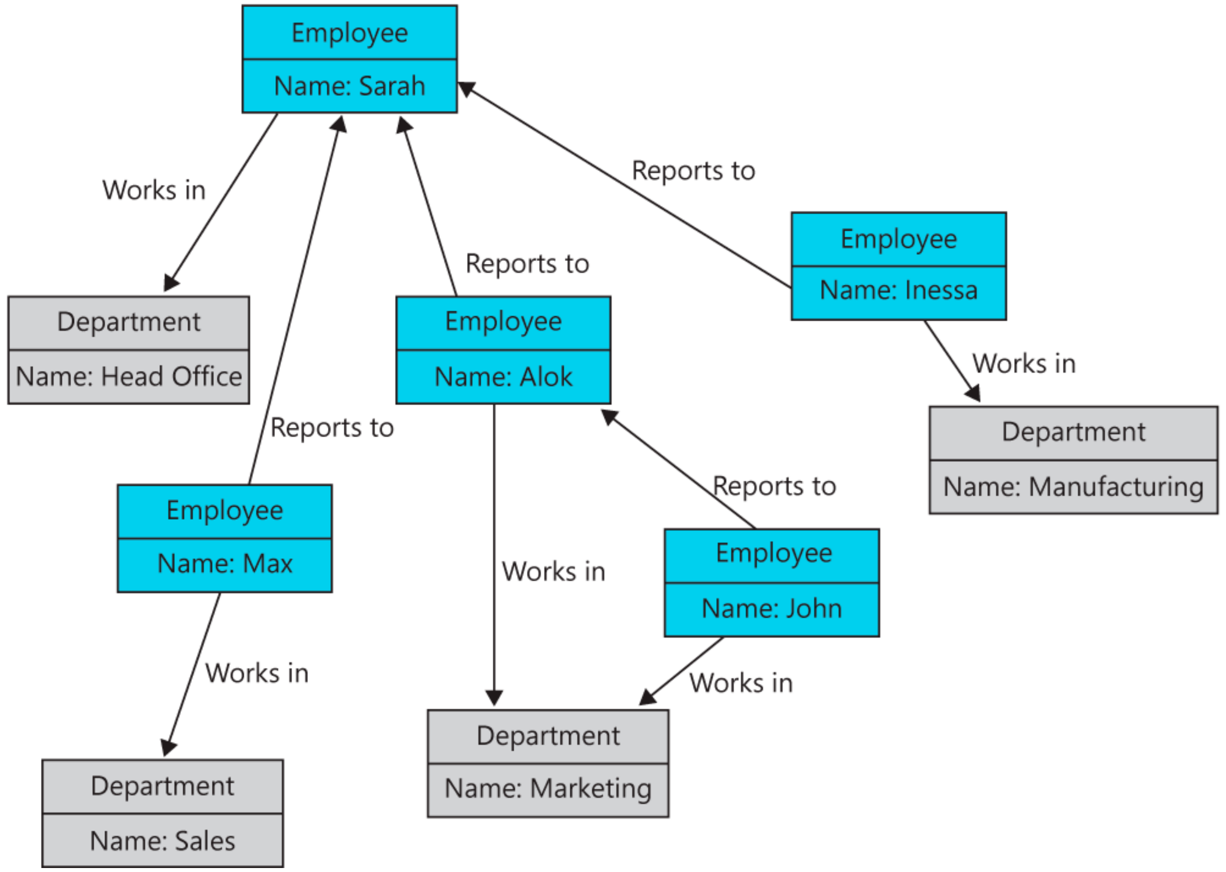
It is conceptually like a key or value store, except that it stores a collection of named fields and data.

Key	Document
1001	{ "CustomerID":99 "OrderItems":[{"ProductID":2010, "Quantity":2, "Cost":520 }, {"ProductID":4365, "Quantity":1, "Cost":18 }], "OrderDate":"04/01/2017" }
1002	{ "CustomerID":220 "OrderItems":[{"ProductID":1825, "Quantity":1, "Cost":120 }], "OrderDate":"05/08/2017" }

- Data fields of a document can be encoded, including XML, YAML, JSON, and BSON.
- Document fields are accessible to the storage management system, allowing applications to query and filter data based on field values.

Graph Databases

A graph database stores two types of information: nodes and edges.



- Nodes store data entities.
- Edges specify the relationships between nodes.
- Both nodes and edges can have properties that provide information about a particular node.
- Edges can also have a direction indicating the nature of the relationship.

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Recommending Database Service Tier Sizing

Azure SQL Database and Azure SQL Managed Instance

The types of service tiers are:

General purpose	Business critical	Hyperscale
Budget-oriented workloads	Low latency workloads	Business workloads

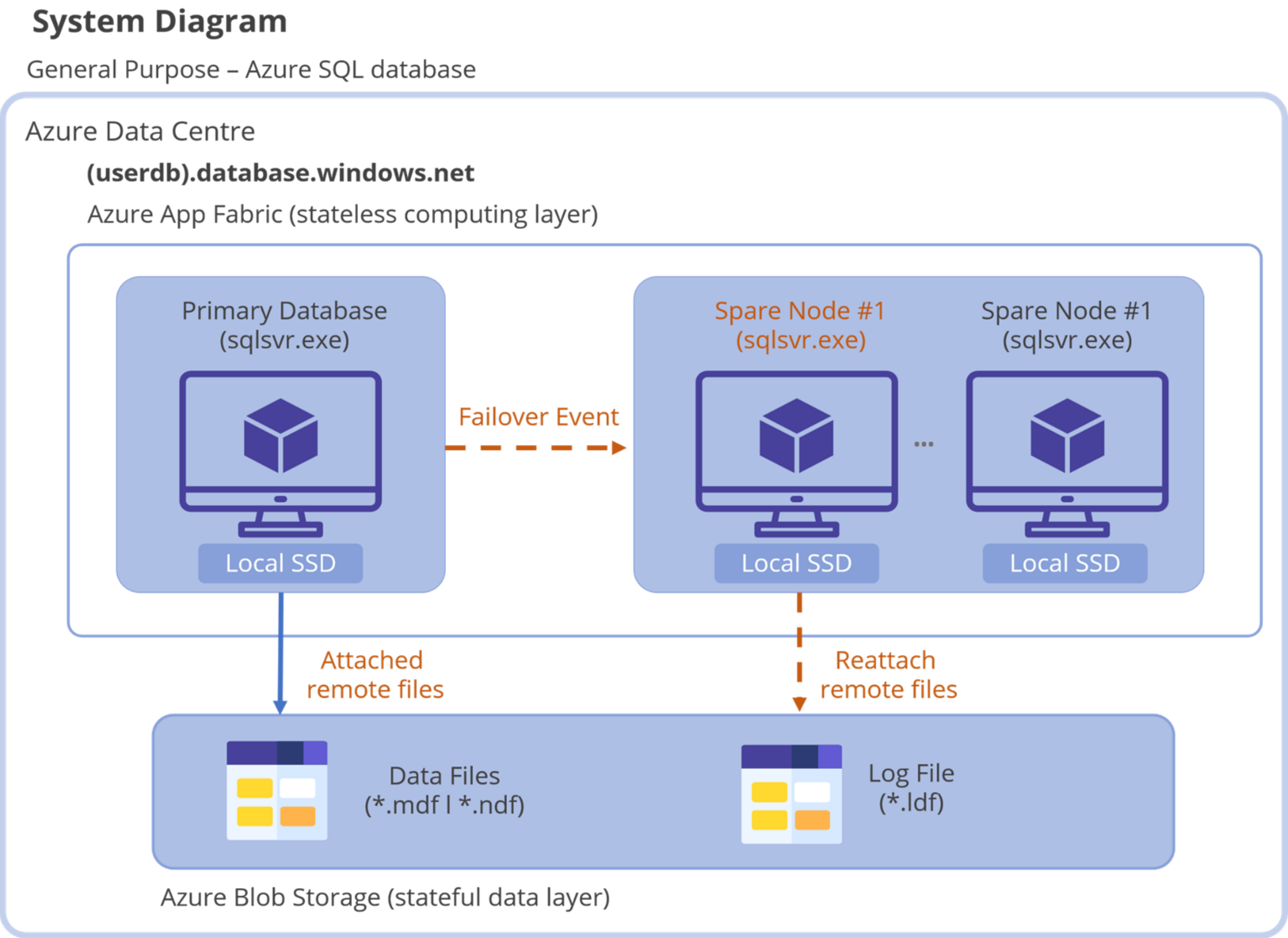
Azure SQL Database and Azure SQL Managed Instance

The following are examples of different usage scenarios:

General purpose	Business critical	Hyperscale
A retail business with moderate traffic requires a cost-effective solution for regular data storage and retrieval, without the need for low-latency access.	An online banking system demands high performance, high availability, and rapid failover to meet stringent operational requirements.	A global customer analytics system processes data from millions of devices or users, needing extensive storage capacity and seamless scalability to accommodate growing data volumes.

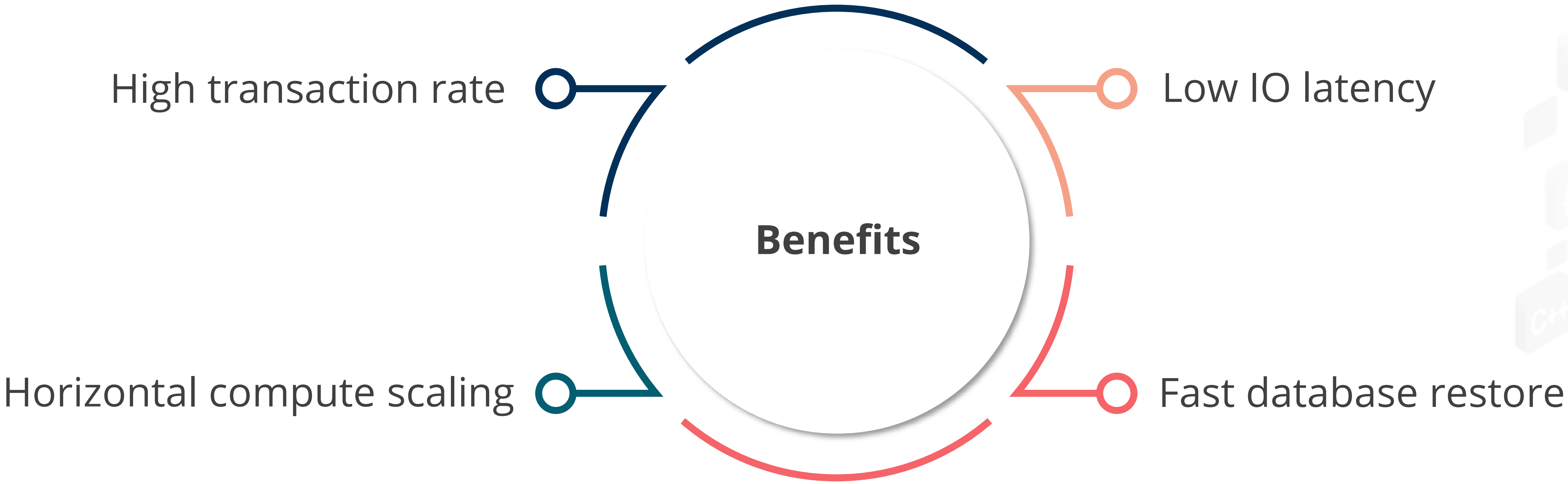
General Purpose Service Tier

It comes with 99% availability.



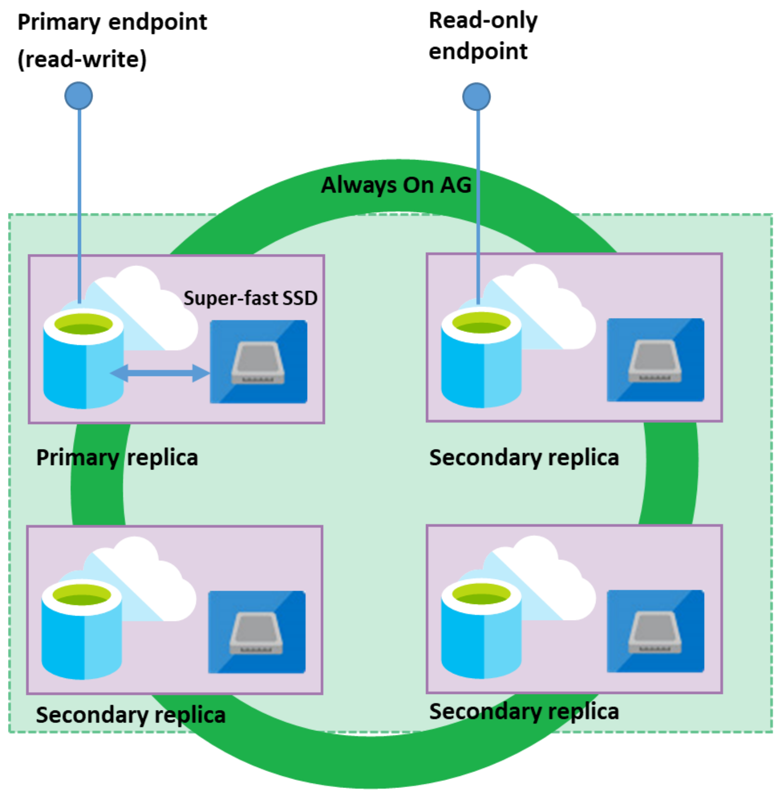
It is a default service tier that is suitable for generic workloads.

General Purpose Service Tier: Benefits



Business Critical Service Tier

It is based on a cluster of database engine processes.



Business Critical service tier: collocated compute and storage

Premium availability is enabled in this service tier.

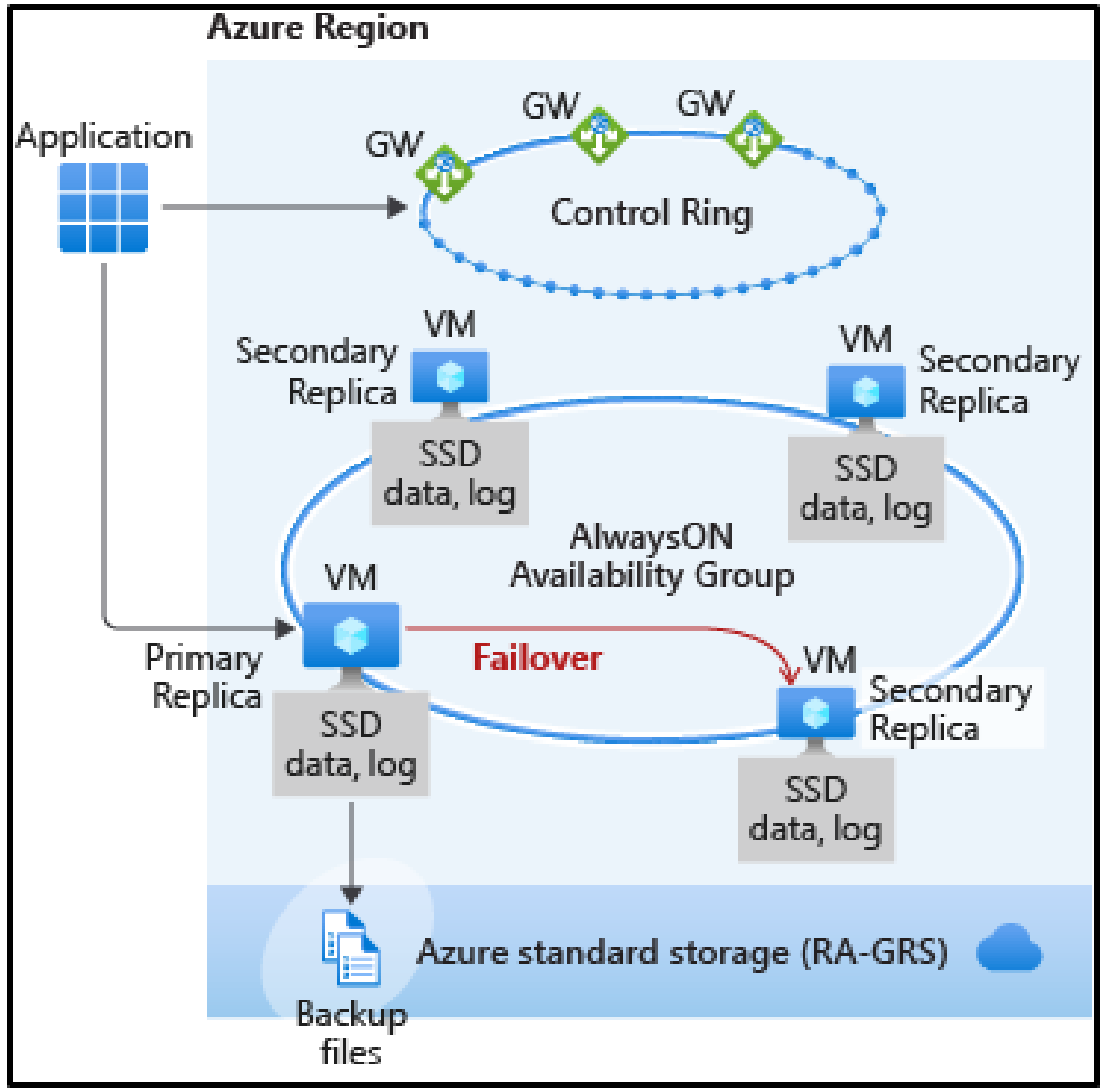
Business Critical Service Tier

It is suitable in situations where there is a need for:

- Minimal latency
- Regular communication between the application and the database
- Increased availability
- Extended transactions with data modifications
- Enhanced resilience and swift recovery from failures
- Substantial time spent on reporting and analytic queries
- Speedy geographical recovery

Business Critical Service Tier Availability

It is created when a four-node cluster is formed.



Source: https://techcommunity.microsoft.com/t5/azure-sql-blog/user-initiated-manual-failover-on-sql-managed-instance/ba-p/1538803

Service Tier Comparison

The difference between the service tiers is given in the table below:

Availability	Resource type	General purpose	Business critical	Hyperscale
Compute size	SQL database	1 to 80 vCores	1 to 80 vCores	1 to 80 vCores
Storage size	SQL database	5 GB - 4 TB	5 GB - 4 TB	Up to 100 TB
TempDB size	SQL managed instance	24 GB per vCore	Upto 4 TB	N/A
In-memory OLTP	SQL managed instance	N/A	Available	N/A
Database size	SQL database	5 GB - 4 TB	Upto 100 TB	5 GB - 4 TB

Service Tier Comparison

General purpose	Hyperscale	Business critical
Least expensive	Less expensive	Most expensive
Latency 5-10 milliseconds	Scales compute resources up and down very quickly	Latency 1-2 milliseconds
99.99 % availability	Instant backups and fast database restores	99.95 % availability
Maximum size 4TB	Maximum size 100TB	Local SSDs on four-node cluster

The service tier characteristics might be different in SQL database and SQL managed instance.
The capability to change from Hyperscale to another service tier is not supported.

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Recommending a Solution for Database Scalability

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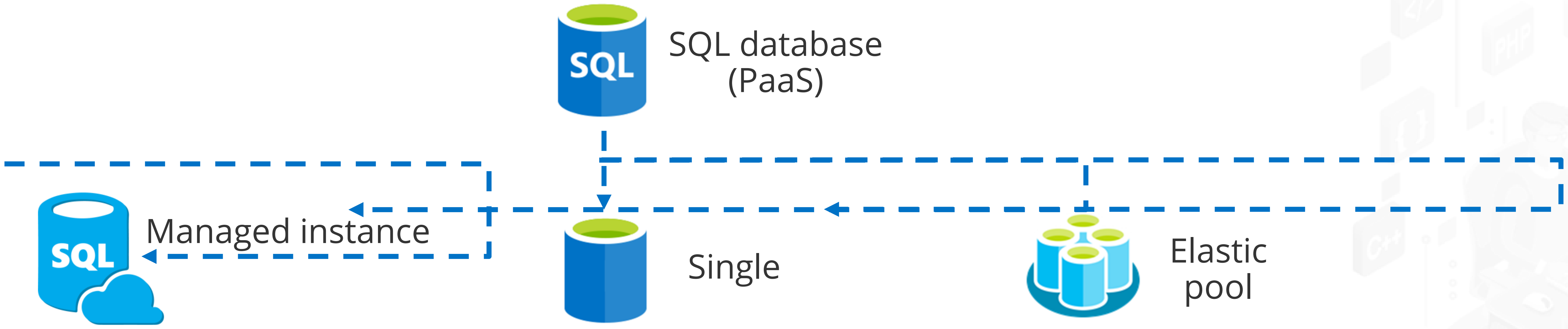
Dynamically Scale Azure SQL Database and Managed Instance

SQL database enables the users to change resources allocated to the databases.



SQL managed instance is a new deployment option that enables frictionless migration for SQL apps and modernization in a fully managed service.

Dynamically Scale Azure SQL Database and Managed Instance

Benefits of scaling

- Optimizes costs
- Enhances versatility
- Mitigates performance issues
- Handles the incoming workload

Azure SQL Database and Mañaged Instance Pricing: Types

It offers two types of purchasing models:

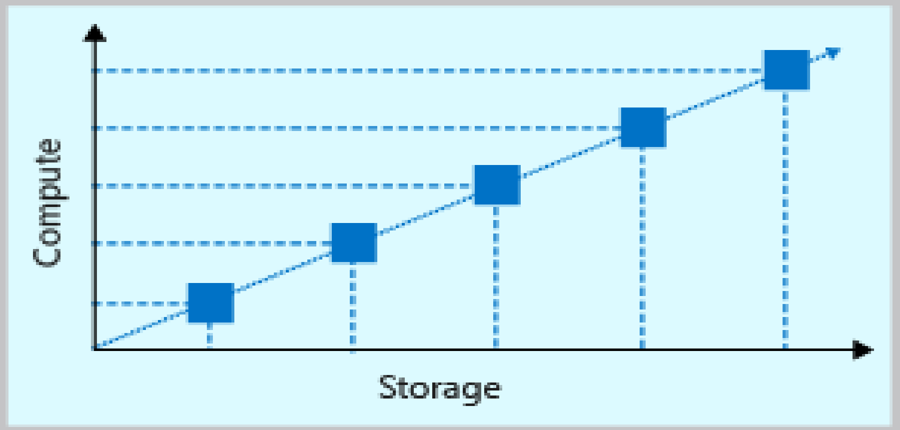
DTU (Database Transaction Unit) based purchasing model	Helps understand the relative resources that are allocated for databases at different compute sizes and service tiers
vCore-based purchasing model	Provides flexibility, control, transparency of individual resource consumption, and a straightforward way to translate on-premises workload requirements to the cloud

DTU-Based Purchasing Model

It is a composite metric that includes CPU, memory, reads, and writes.

DTU model

Simple, Preconfigured



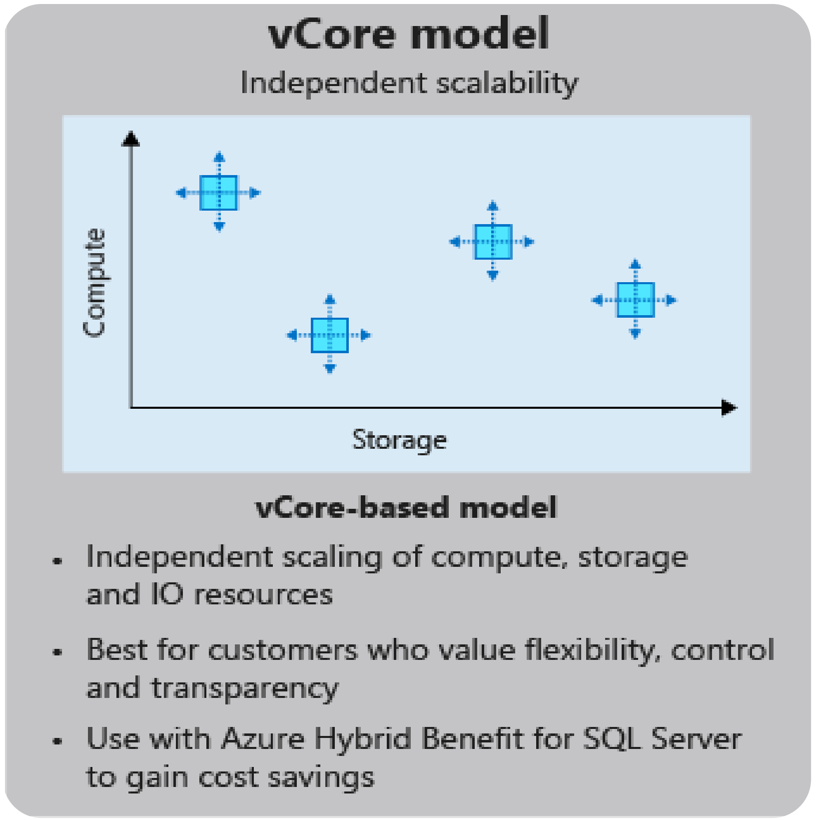
Database Transaction Unit (DTU)-based model

- Bundled measure of compute, storage and IO resources
- Best for customers who want simple, pre-configured resource options

This model provides a list of preconfigured compute resource bundles.

vCore-Based Purchasing Model

It helps in optimizing price.



The vCore-based pricing strategy allows users to choose computing and storage resources individually.

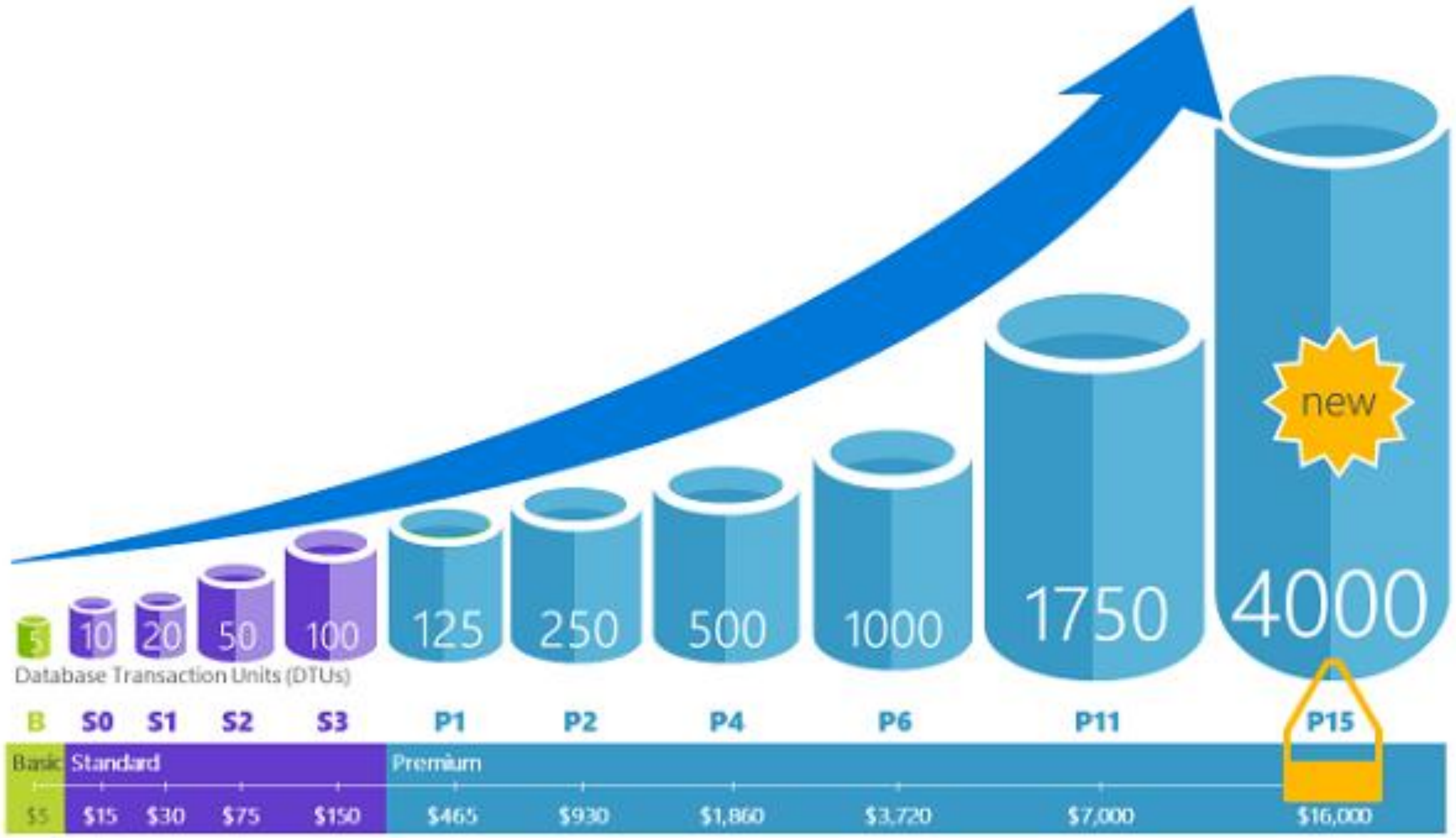
Scaling Azure SQL Database and Managed Instance

Scaling the database can be done via the Azure portal using a slider.

Azure SQL Database offers the DTU-based purchasing model and the vCore-based purchasing model, while Azure SQL Managed Instance offers just the vCore-based purchasing model.

Scale Single Databases in Azure SQL Database

Azure SQL Database allows the users to scale the databases dynamically.



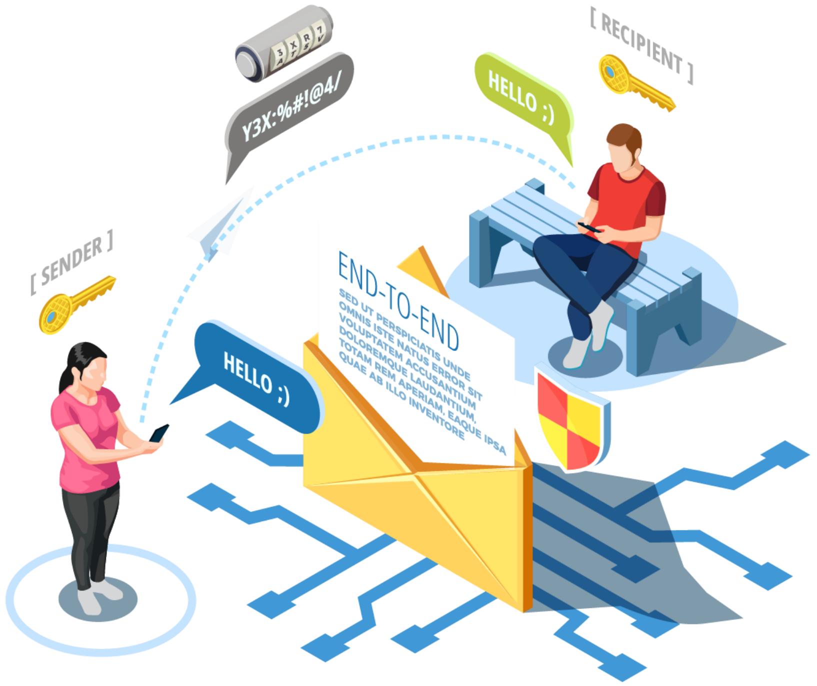
Scale Single Databases in Azure SQL Database

The user can determine the maximum number of resources that will be assigned to each database using either of the purchasing models.

The user can set a maximum resource limit for each group of databases in an elastic pool.

Transport Layer Security (TLS)

It encrypts communication between a client application and an Azure Storage account.



It is a standard cryptographic protocol that assures client and service privacy and data integrity over the Internet.

Transparent Data Encryption (TDE)

It encrypts data files for SQL Server, Azure SQL Database, and Azure Synapse Analytics.

Encrypting data at rest is the terminology for this type of encryption.

Azure Security Audit

The audit methodology relies on Microsoft's recommended best practices for Azure, focusing mainly on:

- Enhancing the Microsoft Defender for Cloud security score for users.
- Employing an industry-recognized benchmark to evaluate an organization's current security posture.

Azure Security Audit: Key Aspects

- Azure conducts internal and external compliance audits on a regular basis, as well as regulatory compliance attestations.
- It examines the policy's specifications.
- It utilizes Azure Governance Visualizer for a comprehensive view of the technical Azure Governance setup.

Creating Azure SQL DB

Duration: 10 Min.

Problem Statement:

You have been assigned a project to construct an Azure SQL DB that will automate most database maintenance tasks such as updating, patching, backups, and monitoring without user involvement.

Outcome:

You will be able to create an Azure SQL Database that automates database maintenance tasks such as updating, patching, backups, and monitoring without requiring user intervention.

Note: Refer to the demo document for detailed steps:
01_Creating_Azure_SQL_DB

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to create an Azure SQL DB:

1. Log in to your Azure portal
2. Search and select Azure SQL
3. Select the SQL database tile
4. Fill in the required fields and create SQL DB

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Design Security for Data

Design Security for Data

Data on the Azure platform is always encrypted, and Azure provides customers with strong data security, both by default and as customer options.

Different states of data

Data **at Rest**	vs.	Data **in Motion**	vs.	Data **in Transit**

Data at Rest

It refers to inactive data stored in databases, hard drives, or devices.

Data **at** **Rest**

Unlike data in transit, it poses lower security risks as it is not actively moving between devices or networks.

Data in Motion

It refers to data that is actively being accessed and processed by users.

Data **in Motion**

It is highly vulnerable during reading, processing, or updating due to its immediate accessibility, exposing it to potential attacks or human errors with serious consequences.

Data in Transit

It refers to actively moving information from one location to another, such as across the internet or through a private network.

Data in Transit

It is less secure than inactive data, making it a prime target for hackers during transmission.

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Design for Azure SQL Edge

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Azure SQL Edge

It is an optimized relational database engine geared for IoT and IoT Edge deployments.

- It provides capabilities to create a high-performance data storage and processing layer for IoT applications.
- It can stream, process, and analyze relational and non-relational data such as JSON, graph, and time-series data.

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Azure SQL Edge

- Provides the same Transact-SQL (T-SQL) programming surface area, making the development of applications or solutions easier
- Facilitates portability among IoT Edge devices, data centers, and the cloud

Advanced Security Features of Azure SQL Edge

Enhancing data security through transparent data encryption and **Always Encrypted** features

Managing access with role-based access control (RBAC) and attribute-based access control (ABAC)

Ensuring compliance with security rules through data classification

Supported Features of Azure SQL Edge

SQL streaming based on the same engine that powers Azure Stream Analytics and provides real-time data streaming capabilities in Azure SQL Edge

The T-SQL function call DATE_BUCKET for Time-Series data analytics

Machine learning capabilities through the ONNX runtime, included with the SQL engine

Unsupported Features of Azure SQL Edge

Database Design	Full-text indexes and search, related DDL commands and Transact-SQL functions, catalog views, and dynamic management views
Database Engine	CLR assemblies, related DDL commands, and Transact-SQL functions, catalog views, and dynamic management views
SQL Server Agent	Subsystems: CmdExec, PowerShell, Queue Reader, SSIS, SSAS, and SSRS
Manageability	SQL Server Utility Control Point

Azure SQL Edge: Editions

Azure SQL Edge is available with two different editions or software plans such as:

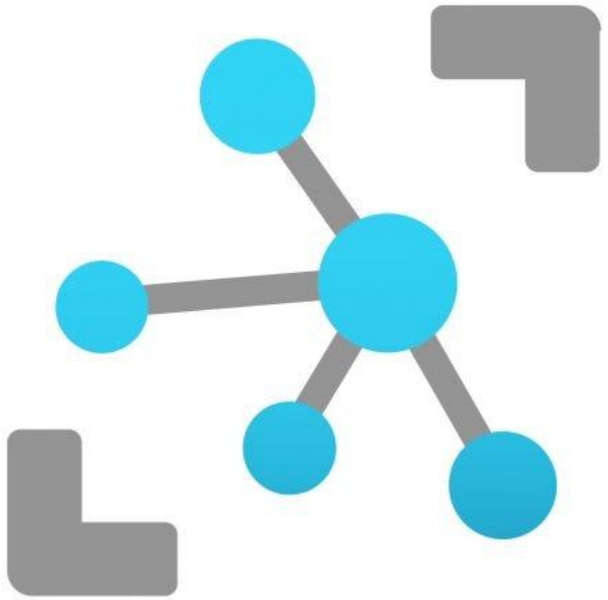
Azure SQL Edge Developer

For development

Azure SQL Edge

For production

Azure SQL Edge: Working



Azure SQL Edge makes designing and maintaining applications easier and more productive. Users can build fantastic apps and solutions for their IoT Edge needs using all their familiar tools and abilities.

Azure SQL Edge: Working

A user can develop in SQL Edge using tools such as:

01 SQL Server Management Studio

02 Azure Data Studio

03 SQL Server Data Tools in Visual Studio

04 Visual Studio Code

TECHNOLOGY

Design for Azure Cosmos DB

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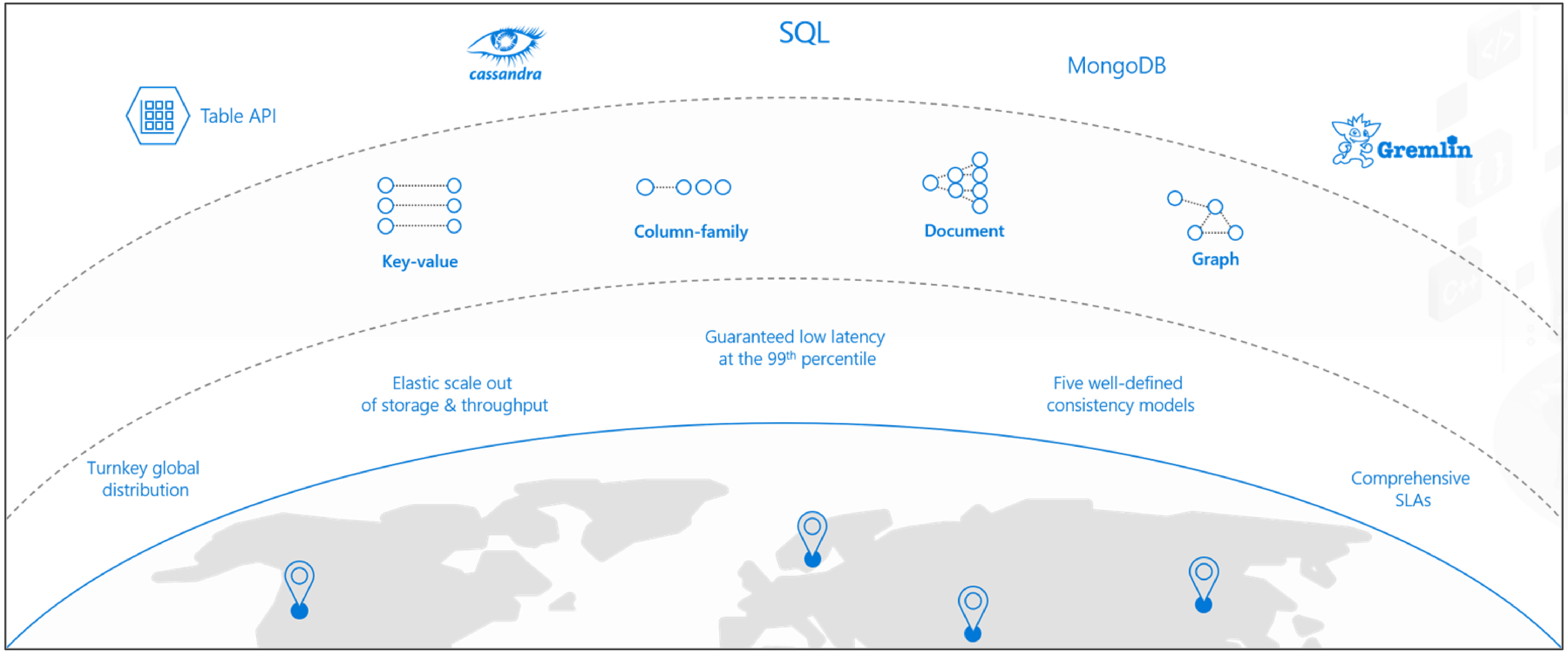
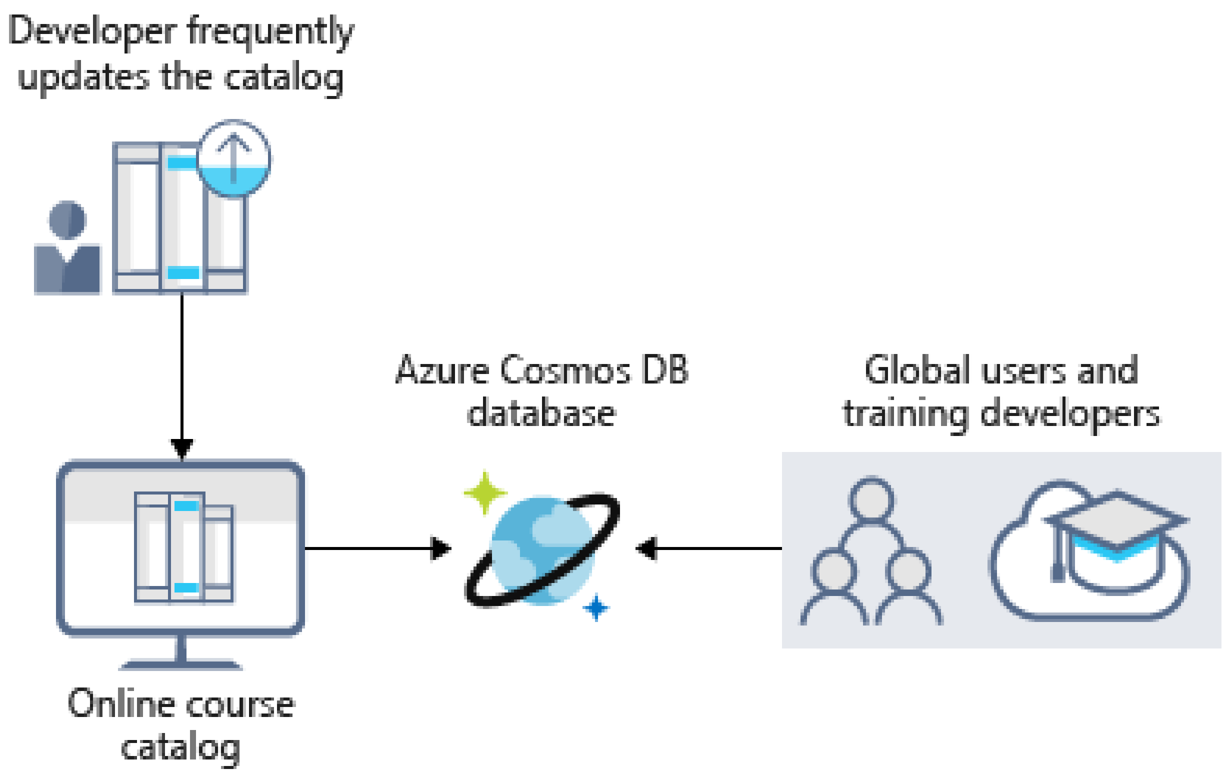
Azure Cosmos DB

It is a globally distributed and elastically scalable database.



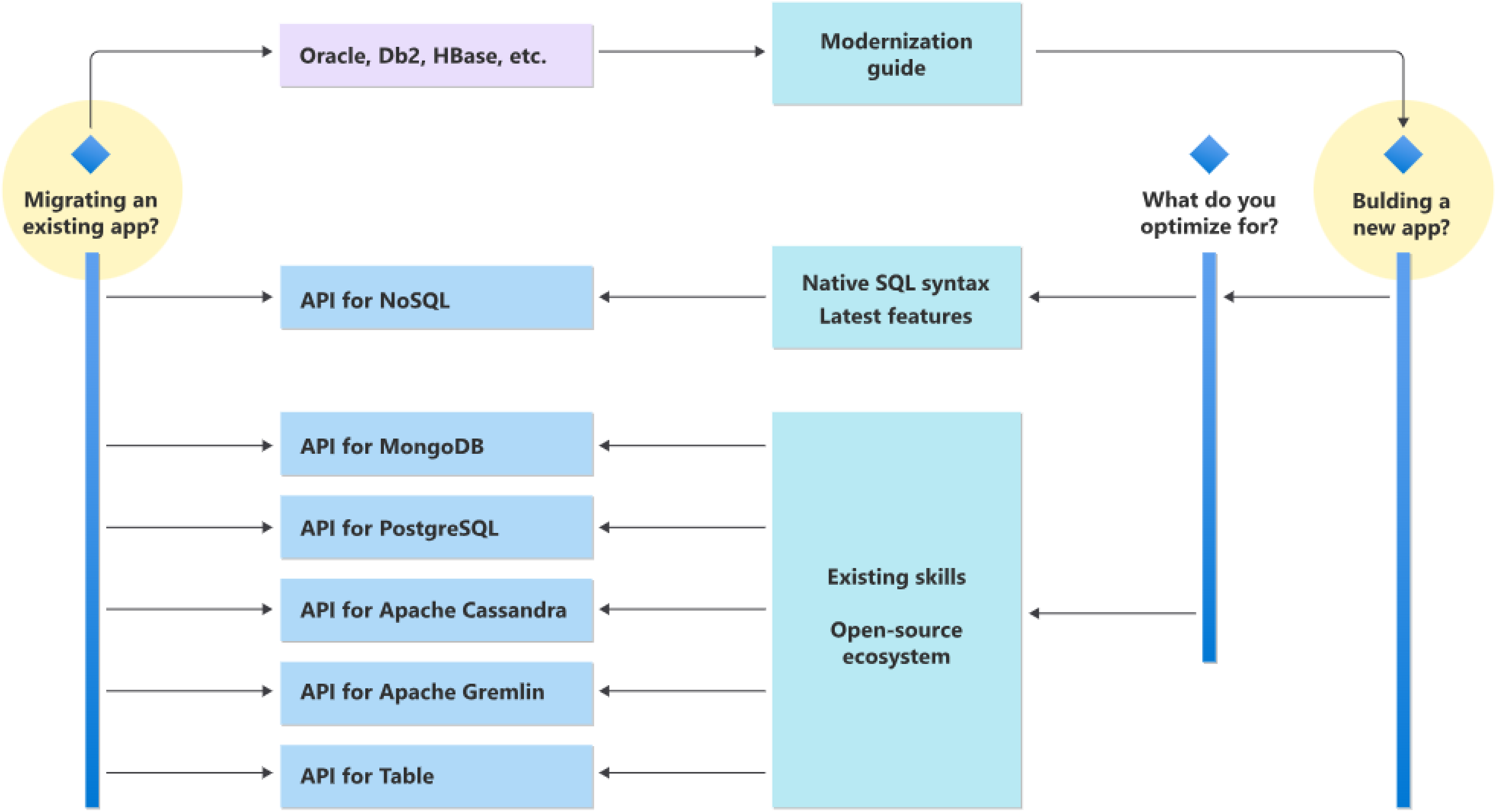
Azure Cosmos DB: Features



- Guaranteed speed at any scale
- Simplified application development
- Mission-critical ready
- Fully managed and cost-effective
- Azure Synapse Link for Azure Cosmos DB
- Azure Cosmos DB is more than an AI database

Database API in Azure Cosmos DB

Decision tree to choose an API in Azure Cosmos DB:



Database API in Azure Cosmos DB

Azure Cosmos DB offers multiple database APIs, such as:

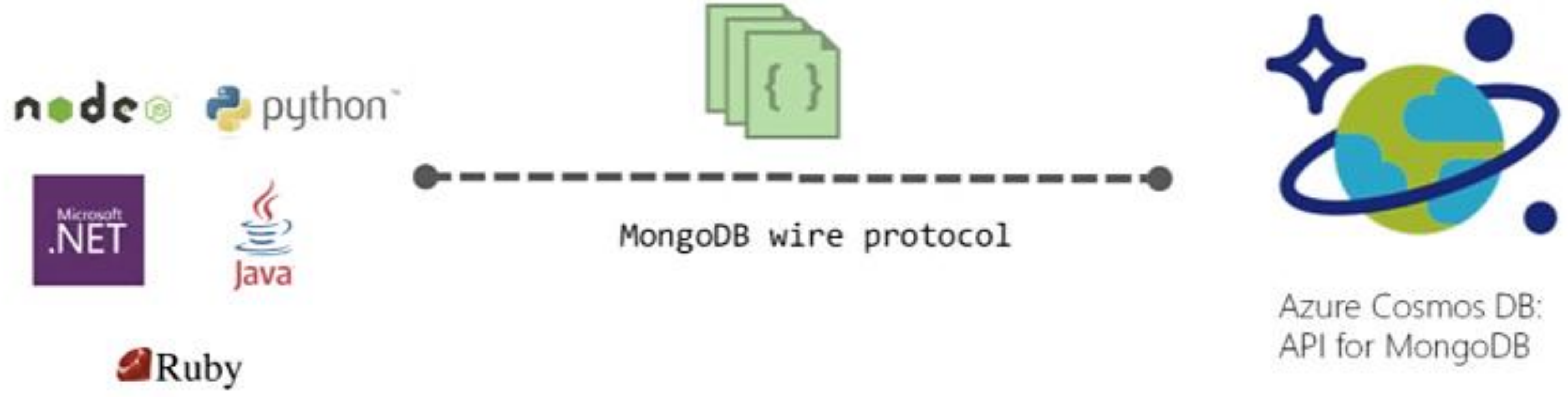
- Core (SQL) API
- API for MongoDB
- Cassandra API
- Gremlin API

Core (SQL) API

This feature being rolled out to Azure Cosmos DB is first available on SQL API accounts.

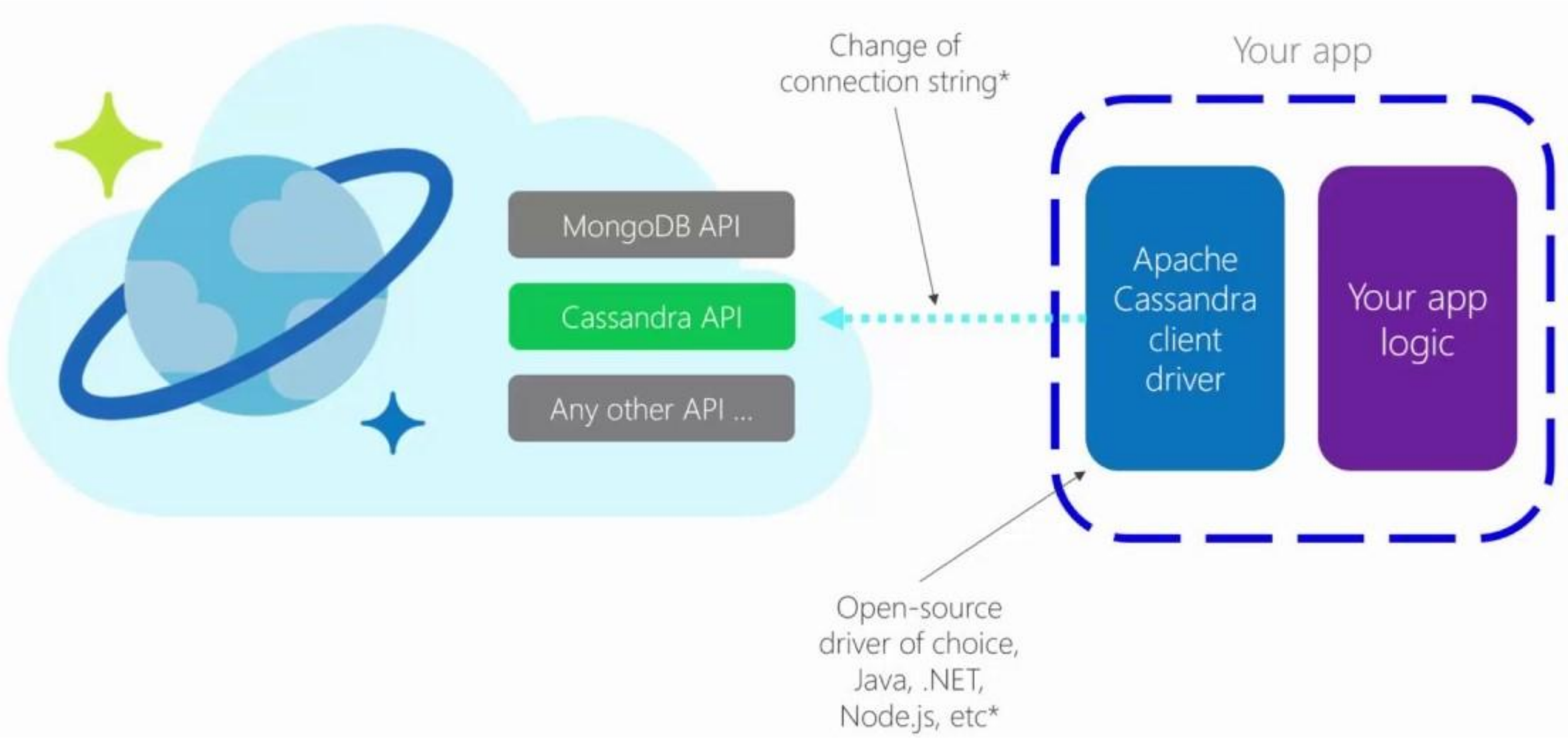
API for MongoDB

The API for MongoDB enables the use of existing client drivers by adhering to the MongoDB wire protocol.



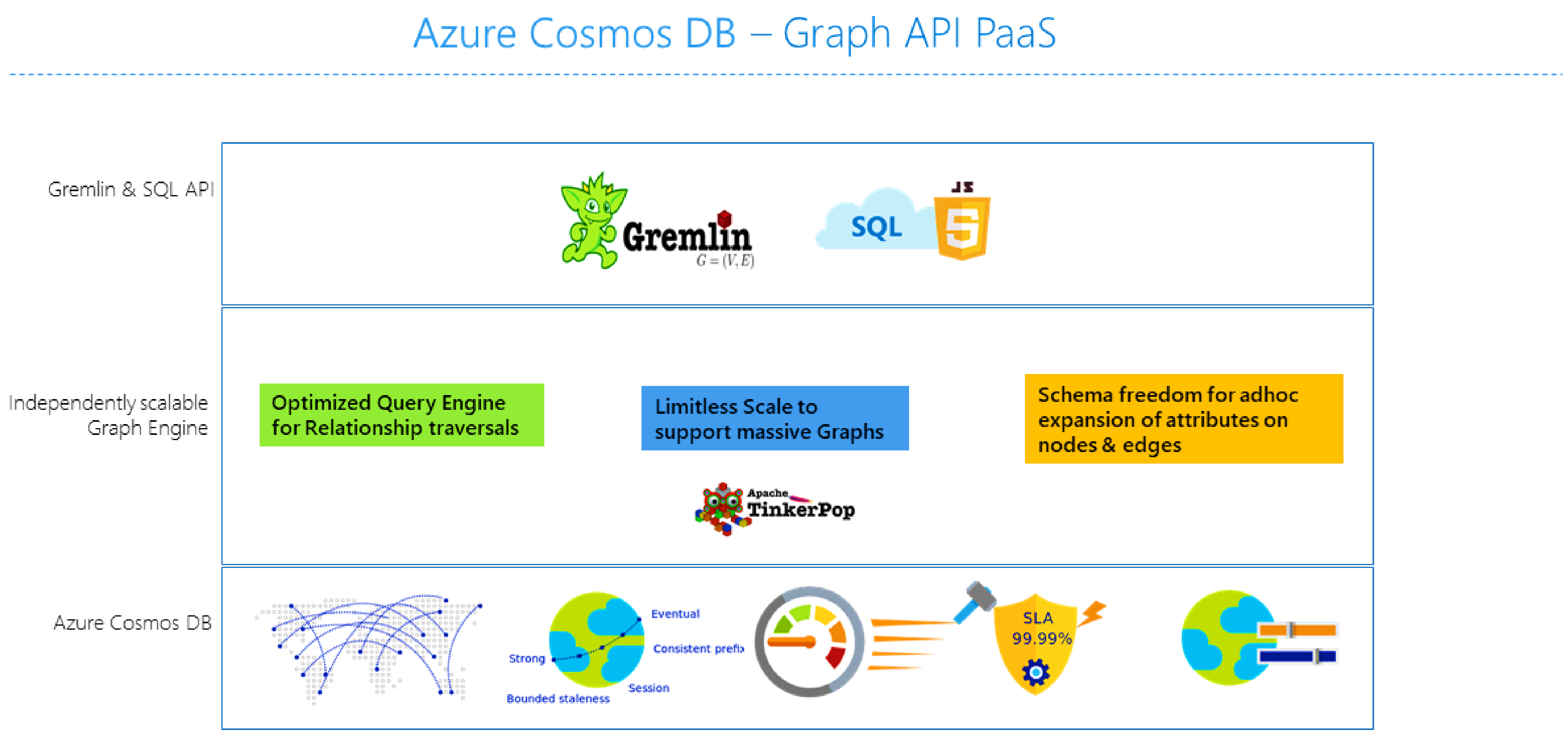
Cassandra API

Cassandra API in Azure Cosmos DB aligns with the philosophy of approaching distributed NoSQL databases, and the API stores data in a column-oriented schema.



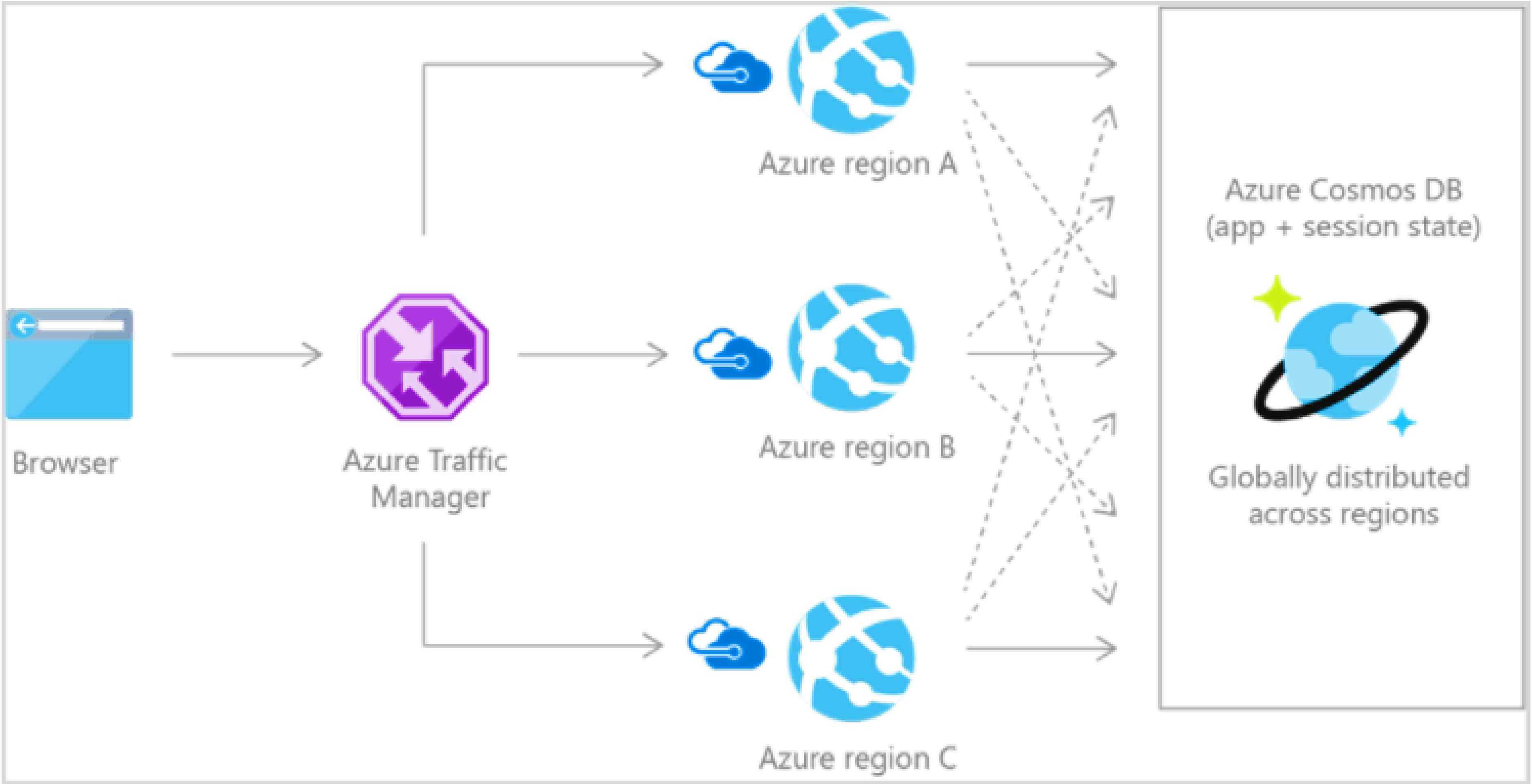
Gremlin API

Azure Cosmos DB's Gremlin API combines the power of graph database algorithms with highly scalable, managed infrastructure.



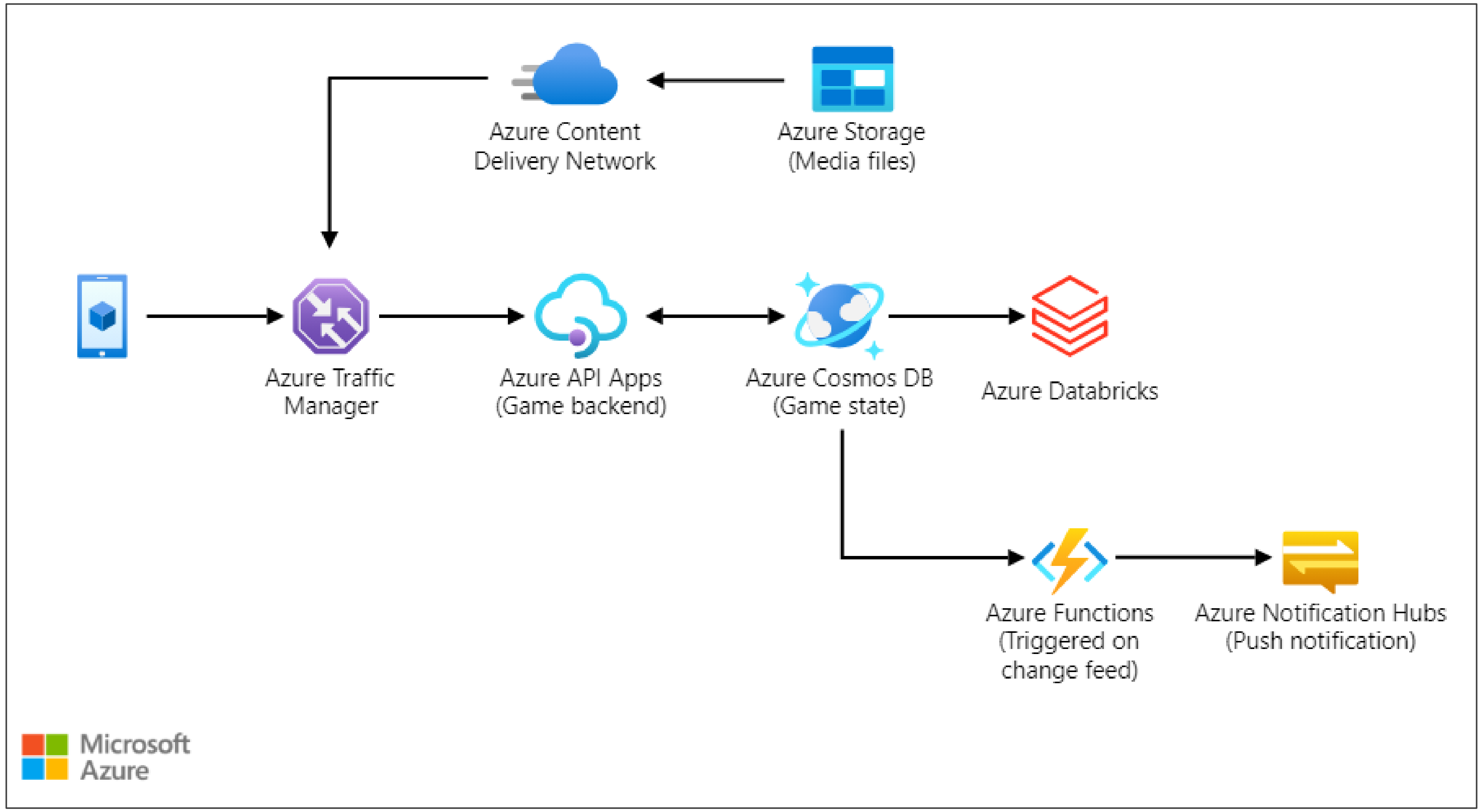
Use Case of Azure Cosmos DB

Applications that integrate with third-party social networks must respond to changing schemas from these networks.



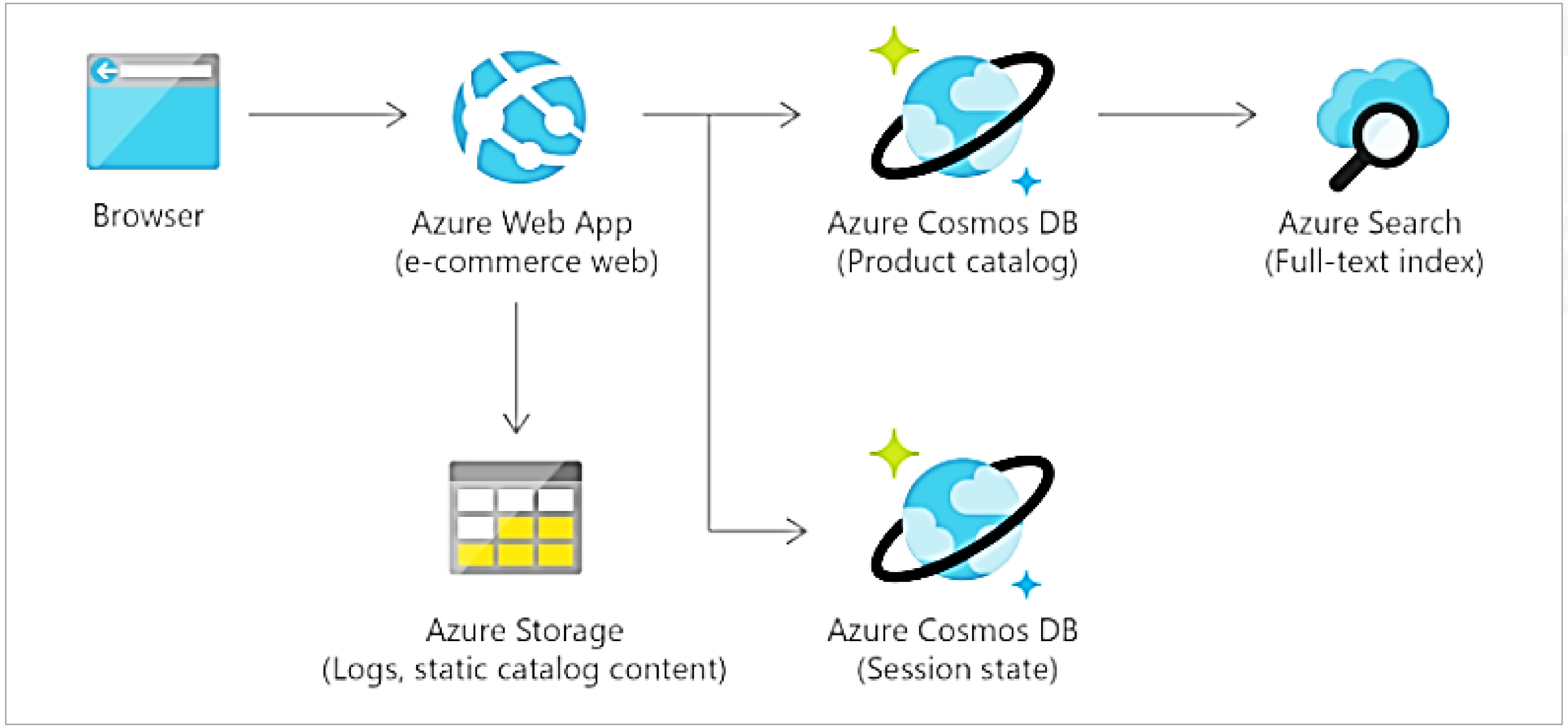
Use Case of Azure Cosmos DB

Modern games perform graphical processing on mobile or console clients but rely on the cloud for in-game content.



Use Case of Azure Cosmos DB

Microsoft's e-commerce sites, such as the Windows Store and Xbox Live, rely heavily on Azure Cosmos DB.



Azure Cosmos DB Table

Features	Azure Table storage	Azure Cosmos DB Table API
Latency	Fast but no upper bounds on latency	Single-digit millisecond latency for reads and writes
Query	Execution uses index for primary key and scans otherwise	Queries that can take advantage of automated indexing on properties
Indexing	Only primary index on Partition Key and Row Key	Automatic indexing on all properties by default without index management
Global distribution	Single region with one optional readable secondary read region for high availability	Turnkey global distribution from one to any number of regions

Azure Cosmos DB Table

Features	Azure Table storage	Azure Cosmos DB Table API
Throughput	Scalability limit of 20,000 operations per second	Highly scalable with dedicated reserved throughput per table that is backed by SLAs and Accounts with no upper limit
Consistency	Strong within primary region and Eventual within secondary region	Five well-defined consistency levels to trade off availability, latency, throughput, and consistency based on needs.
SLAs	99.99% availability, depending on the replication strategy	Comprehensive SLAs covering availability, latency, throughput, and consistency.
Pricing	Consumption-based	Available in both consumption-based and provisioned capacity modes

Managing Cosmos DB

Duration: 10 Min.

ASSISTED PRACTICE

Problem Statement:

You need to create a Cosmos DB account and identify different options for deploying and configuring Cosmos DB. You also need to add some data to the Cosmos DB to check its functionality further by running queries.

Outcome:

You will be able to create and configure a Cosmos DB account, deploy various options, add data, and run queries to test its functionality.

> **Note:** Refer to the demo document for detailed steps:
> 02_Managing_Cosmos_DB

Assisted Practice: Guidelines

Steps to manage storage access keys:

1. Create a Cosmos DB account
2. Create a database
3. Add some data to the cosmos DB database
4. Query the data using SQL APIs

Key Takeaways

- Azure SQL Database is a fully managed platform as a service (PaaS) database engine.
- RDBMS implements a transactionally consistent mechanism that confirms to the ACID model for updating information.
- Transport Layer Security(TLS) encrypts communication between a client application and an Azure Storage account.
- Azure SQL Edge is an optimized relational database engine geared for IoT and IoT Edge deployments.

TECHNOLOGY

Knowledge Check

Knowledge Check 1

What are the different database systems used for SQL and NoSQL databases?

A. Relational database management systems

B. Key or value stores

C. Graph databases

D. All the above

Knowledge Check 1

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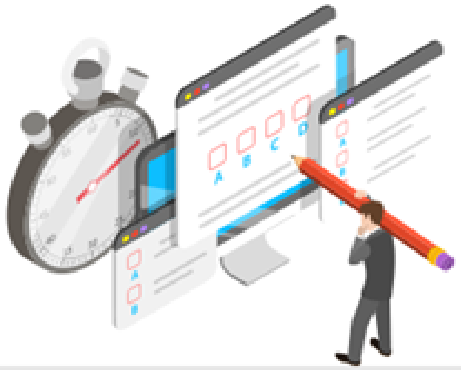

The correct answer is **D**

All of the above are different database systems used for SQL and NoSQL databases.

Knowledge Check 2	**Which service tier is based on a cluster of database engine processes?**

A. Business critical

B. General purpose

C. Hyperscale

D. None of the above

Knowledge Check
2

Which service tier is based on a cluster of database engine processes?

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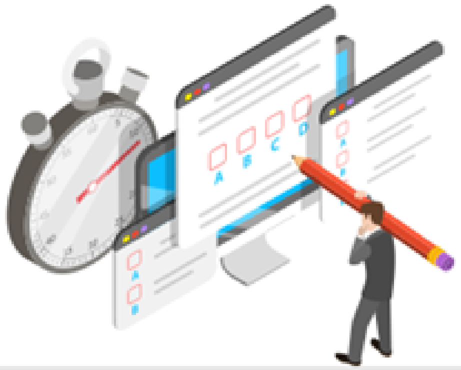
D. None of the above



The correct answer is **A**

Business critical service tier is based on a cluster of database engine processes.

Knowledge Check
3

Which type of encryption uses different keys to encrypt and decrypt?

A. Symmetric

B. Asymmetric

C. Transitive

D. Distributive

Knowledge Check 3

Which type of encryption uses different keys to encrypt and decrypt?

A. Symmetric

B. Asymmetric

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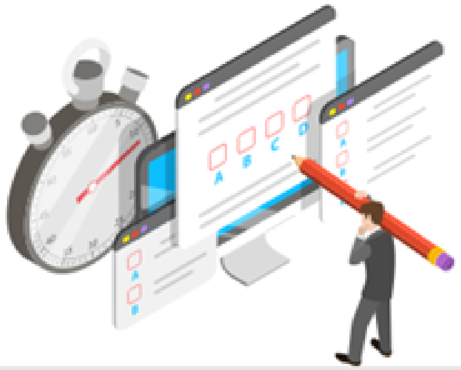
D. Distributive



The correct answer is **B**

Asymmetric encryption uses a public and a private key.

Knowledge Check 4	Data Encryption in Azure storage depends upon which of the following?

A. Performance tier

B. Access tier

C. Deployment model

D. None of the above

Knowledge Check
4

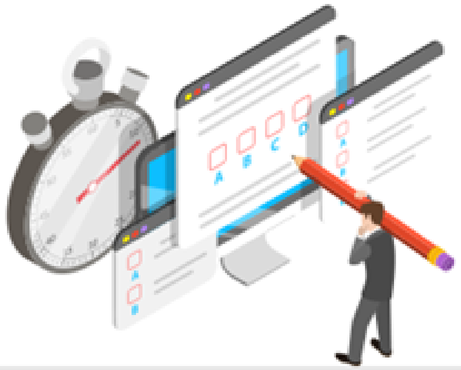
Data Encryption in Azure storage depends upon which of the following?

A. Performance tier

B. Access tier

C. Deployment model

D. None of the above



The correct answer is **D**

Data in a storage account is encrypted regardless of performance tier (standard or premium), access tier (hot or cool), or deployment model (Azure Resource Manager or classic).

Knowledge Check

5

Which of the following platform is supported by Azure Storage Explorer?

A. Web

B. Windows

C. OSX

D. Both B and C

Knowledge Check
5

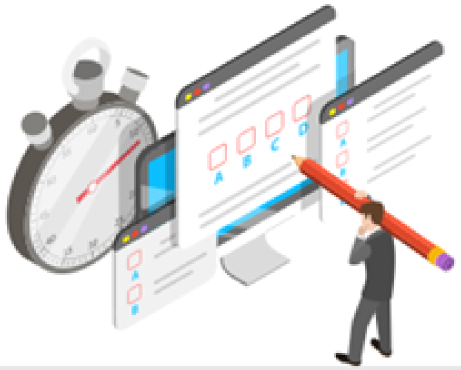
Which of the following platform is supported by Azure Storage Explorer?

A. Web

B. Windows

C. OSX

D. Both B and C



The correct answer is **D**

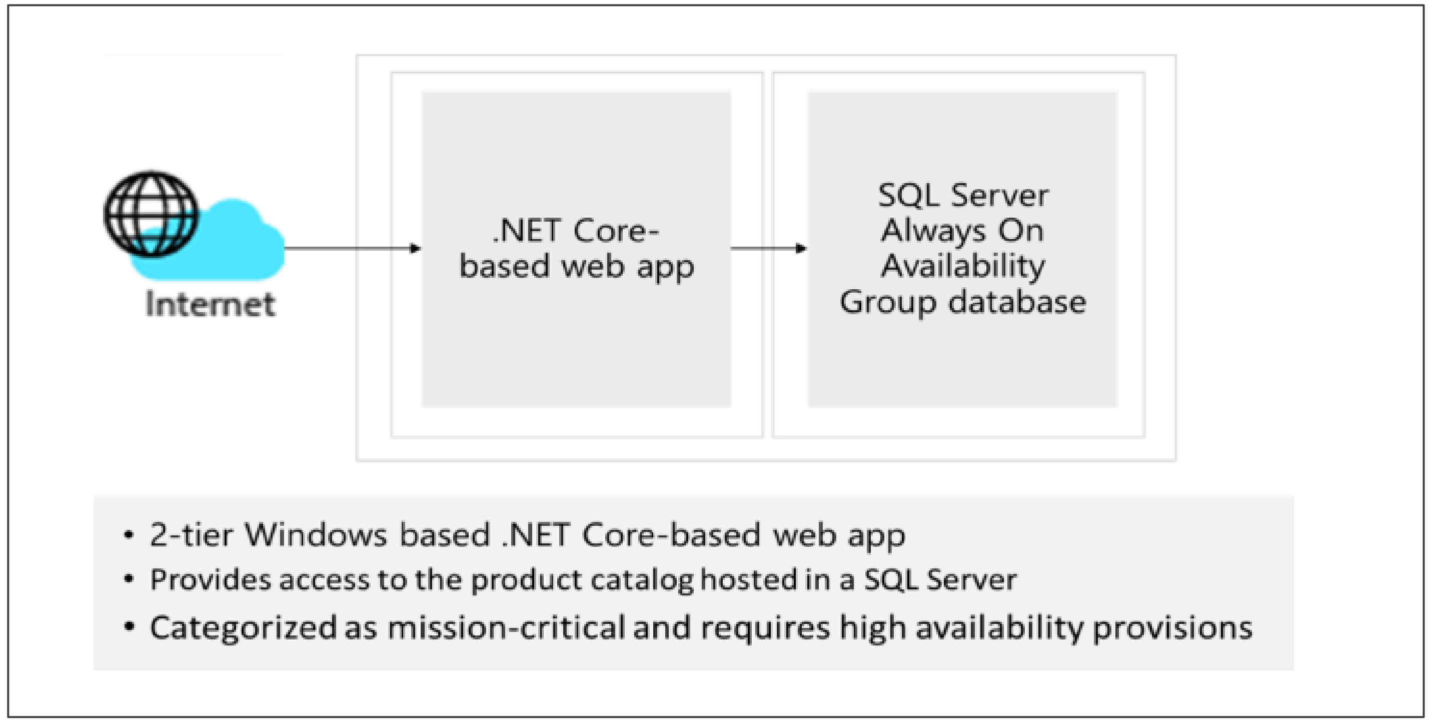
Azure Storage Explorer supports windows and OSX platform.

Case Study

Design for Relational Data Storage Solutions

Scenario

BritTech Solutions is looking to move its existing public website database into Azure, as the website front end is being moved there as well. The website front end will initially only be deployed in two regions for redundancy. However, it is expected that as traffic increases, the website will be replicated in other regions around the world. The database, which you are being asked to migrate, holds the product catalog and all online orders. Currently, the database runs on a single Microsoft SQL Server Always On availability group on-premises.



Design for Relational Data Storage Solutions

Scenario

Primary concerns of Tech Solutions Inc.:

- **High availability**: A primary concern for BritTech Solutions is that this database be highly available, as it is critical to their business. Any outages may result in lost sales or customer confidence.

- **Website performance**: While the performance of placing orders is satisfactory, browsing or searching pages with many items listed is reported as sluggish.

- **Security**: BritTech Solutions is very concerned about personal or financial information stored in the database being exposed. In addition to implementing proper security measures, the security team must verify that industry-standard best practices are implemented when possible.

Design for Relational Data Storage Solutions

Requirements

- Design the database solution. Your design should include authorization, authentication, pricing, performance, and high availability.
- Diagram what you decide and explain your solution.

How are you incorporating the well-architected framework pillars to produce a high-quality, stable, and efficient cloud architecture?

Designing Solution for Database

Duration: 10 min.

Project Agenda: Design a solution for database

Description: You have been given a project to move a database from on-premise to the cloud. The data is in a document format. You need to choose the right database option. You also need to ensure that you have support for querying items using the Structured Query Language (SQL) syntax. Suggest a solution for this migration.

Perform the following:

1. Create an Azure Cosmos DB account by selecting Core (SQL) API option.

LESSON-END PROJECT